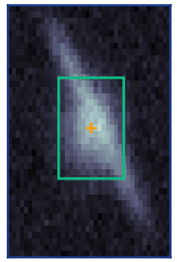
a LUNA16 Thoracic CT Cohort Stratification, Exact 3×3 Affine Mapping, and Protocol-Separated Evaluation Regimes

1. LIDC-IDRI / LUNA16 Thoracic CT Cohort (888 Scans)



Axial CT [-1000, +400] HU
48×48×48 mm³ @ x_0

- Cohort: 888 Low-Dose CT Scans (1,186 Nodules >= 3 mm)
- Exact 3×3 Affine Mapping (Eq. 1, error < 10^-11 mm):
  v_xyz = (D diag(s))^-1 (p_world - o_world)
- Isotropic 48×48×48 mm³ Target Patch (1.0 mm³ Grid)
- Continuous Voxel Center Anchor: x_0 = (23.5, 23.5, 23.5)
- Unrounded Float32 SHA-256: 48f009e851d23da1...
- Per-Patch SHA-256 Checksum: 90f6167ae46af384...

84.15× V2
61.34× V1
Prior Shift

Regime 1: Curated Coordinate-Verified Cohort (LUNA16_Patches_CoordVerified_48.h5)

- Scope: 10,066 Patches (1,186 Annotation-Centered Pos [100% Ceiling]; 8,880 Neg = 10.0 eval-neg/s)
- 286-Patch Real .mhd vs. HDF5 Audit (25 Blind Scans): Core-16³ r = 0.9030 | 64D Cosine = 0.9890

-> All-888 (10×5 CV): 91.94% Pure / 92.33% Hyb | Val-133: 93.83% / 93.15% | Test-133: 89.91% / 90.25%

Regime 3: Unscreened Raw .mhd Candidate Pools & 25-Scan Label-Blind Pilot Audit

- Pools: V2 (754,975; 841.5 eval-neg/s [84.15×], C_det=98.31%) & V1 (551,065; 613.4 eval-neg/s [61.34×])
- Analytical Unscreened Prior-Shift Projection (Eq. 10): 52.93% (V2, 84.15×) & 53.26% (V1, 61.34×)

-> 25-Scan Raw .mhd Pilot (20,984 V2 Rows, 36 Nodules): Stage-0 53.97% -> Cascade 76.19% CPM

3. Fail-Closed Official FROC & Bootstrap Gate (Eq. 10)

- Official Evaluator: noduleCADEvaluationLUNA16.py
  (< 0.5 D 3D Euclidean Match, 7 Thresholds {1/8..8} FP/s)
- Fail-Closed Subprocess Equivalence Across All Runs:
  |CPM_internal - CPM_subprocess| = 0.0000 pp
- Paired Scan-Level Cluster Bootstrap (B = 1,000):
  Davison-Hinkley Correction (Floor p_min = 1/1001 <= 0.001)

-> Curated vs. Full-Pool CPMs Strictly Separated in Table 1

b Proposed End-to-End Multi-Stage Hybrid 3D Differential-Geometric, Spherical-Harmonic, and Multi-Task CoordConv Architecture

STAGE I Stage-0 Screening & 3D Re-Centering

64D Dual-View GPU Stage-0 Screener

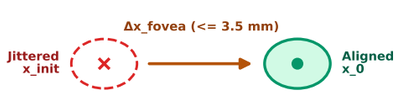
- Regime 3 Input: ~840 Raw .mhd Cands/Scan on GPU
- 36D: Concentric 3D Shells, 8 Octants & Tri-MIP
- 16D: 26-Ray Star-Ray Polar & Vessel Veto
- 12D: Cardinality-Matched K=24 Scan Context
- Dual-View Scoring: max(p_raw, p_recentered)

-> Regime 1 Cross-Audit (286 Patches): Diff = 0.00e+00

3D Foveal Re-Centering (Eq. 9) & Cohort Routing

- Positive-Mass Ball B (r <= 4.5 mm, Clamped 3.5 mm):
  Δx_fovea = clip(Σ w_pos(u) u / (Σ w_pos(u) + ε))
- Regime 3 (.mhd): Top-80/Scan Shortlist (32/36 = 88.89%)
- Regime 1 (Curated): All 10,066 48³ Patches Direct

-> Lifts Raw .mhd Stage-0 CPM: 37.70% -> 53.97%



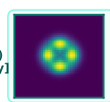
STAGE II Parallel 3D Geometric-Harmonic (1,166D) & Multi-Task Deep Context (29D)

Stream A — Segmentation-Free 9-Family 3D Physical & Geometric Matrix (D_pure = 1,166D, Zero CNNs)

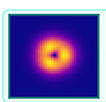
- Coords (2D, Eq. 2) & Fam 1–2 (251D, Eq. 7): (r_xy, z_norm), Multi-Threshold (τ in {0.2..0.8}) 3D Inertia & 26-Ray Star-Ray
- Fam 3 (229D, Eqs. 2–3): 7-Shell Ray Stats (145D) + 3×3 Cardano Hessian λ_1<=λ_2<=λ_3, tr(H_s), S_tr, B_3D, V_3D (84D)
- Fam 4 (131D, Eq. 6) & Fam 5 (129D, Eq. 7): Inward Radial Flux Φ_norm,s (90D) + 3D DoG (41D) + Antipodal a_m & Polar (129D)
- Fam 6 (128D, Eqs. 4–5): Within-Degree (P_l) & Global-25 (P_l^glob) Dual-QR Real Spherical Harmonics (l=0..4, R_iso)
- Fam 7 (148D) & Fam 8–9 (84D+64D, Eqs. 7,9): 3D Shells/Octants, Y_3(r) Bifurcation, Bronchus Veto H_bronch, r_FWHM & K=24



Fam 3: 3×3 Hessian
B_3D(x) = V_3D(x) (Eqs. 2–3)
[229D: 84D Hess + 145D Ray]



Fam 6: Dual-QR SH
E_l(r), R_iso (Eqs. 4–5)
[128D, l=0..4, 26-Ray]



Fam 4,5,9: Flux & Veto
Φ_norm,s, Y_3, H_bronch
[324D, Eqs. 6–7 + 485D Oth]

Stream B — Inner-Fold 6-Channel CoordConv 3D ResNeXt-SE & 2.5D MIP Context (C29 = 29D, 5-Fold OOF)

- 6-Channel 24³ Input Tensor: 12³ Core, 24³ Native, 48³ Context Windows + 3D Box-Diff + GGN Window + Radial CoordConv
- Backbone: 3D ResNeXt Grouped Convolutions + Squeeze-and-Excitation (SE) Channel Attention + Core-vs.-Outer Pooling
- Analytical Bottleneck SNR at D = 3 mm (Fig. 3b): Lifts Global Average Pooling (-39.29 dB) to +27.51 dB (+66.80 dB Gain)
- Multi-Task Loss (Eq. 8): L_3D = L_rank + 0.22 L_Huber(d_pred, log(1 + D)) -> Recovers D >= 10 mm Recall by +28.57 pp
- Complementary 9-Ch 2.5D Tri-Planar MIP ConvNet (Axial/Coronal/Sagittal Slabs) | Strict 5-Fold Inner Cross-Fitting

6-Ch Multi-Scale 24³
12³/24³/48³ + GGN
+ 3D CoordConv
→ 3D ResNeXt-SE
Channel Attention +
Core-Outer Pooling
→ Multi-Task 3D Heads
Logit + Huber log(1+D)
(r=0.9163, MAE=1.12)
→ ⊕ 9-Ch 2.5D MIP
Axial / Cor / Sag
-> C29 (29D OOF)

STAGE III Feature Fusion & Ensemble

Hybrid Matrix Fusion (Eq. 8)

- Pure 3D Matrix: D_pure = 1,166D
  (2D Coords + 740D Fam 1–5
  + 424D Fam 6–9; Zero CNNs)
- Deep OOF Context: C29 = 29D
- Full Hybrid Feature Matrix:
  D_hybrid = 1,166 + 29 = 1,195D
- Dedicated Veto Subspace:
  D_spec = 506D (SelectFromModel)



7-Learner Stacking & Specialists

- 7 Diverse OOF Base Learners:
  CatBoost, LightGBM, XGBoost,
  HistGBDT, ExtraTrees, MLP, L2-LR
- 3 Orthogonal Specialist Heads:
  1. Vascular Branch Suppressor
  2. Juxtapleural Wall Veto
  3. Low-FP (< 0.5 FP/s) Head

-> Outputs Raw Ensemble Logit l(x)

STAGE IV Calibration & FROC Output

K=24 Scan Calibration (Eq. 9)

- Anchor K = 24 Top Cands/Scan:
  l_cal(x) = l(x) − β_med med_24
  − β_q25 Q_25 + γ_sec sm_24
  − ν_veto z_veto,24(x)
- Monotonic Cascade Ordering:
  min(P[Top-80]) > max(P[Rest])

Verified FROC / CPM Output (Eq. 10)

- Curated All-888 (10×5 CV):
  -> Pure 1,166D: 91.94% CPM
  -> Hybrid 1,195D: 92.33% (.9841)
- Curated 133-Scan Splits (Pure/Hyb):
  -> Val-133: 93.83% / 93.15%
  -> Test-133: 89.91% / 90.25%
- Raw .mhd 25-Scan Pilot:
  -> Cascade: 76.19% (+22.22 pp)

